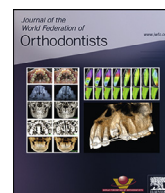




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Research Article

Towards automated model analysis: A multiview AI segmentation of 3D dental scans

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ABSTRACT

Objective: Precise identification of tooth and gingival boundaries in digital models is essential for effective orthodontic diagnosis, treatment planning, and appliance fabrication. Recent advances in artificial intelligence (AI) offer opportunities to automate this critical step with accuracy and generalizability.

The objective of this study is to develop and evaluate an AI-based multiview segmentation approach that reduces manual workload, and supports daily orthodontic workflows using 3D intraoral scans.

Methods: A total of 1200 dental models from the public 3D Teeth Challenge dataset and 29 clinical 3D intraoral scans were used. Each 3D model was converted into multiple 2D images using a modified multiview approach. An AI segmentation model, based on the Mask2Former architecture, was trained to segment tooth boundaries automatically. Performance was assessed using mean Intersection over Union (mIoU) and Dice Similarity Coefficient (DICE) scores, comparing results to existing approaches.

Results: The proposed model achieved high accuracy on both public and clinical datasets. On the public testing dataset, it reached an mIoU score of $93.1\% \pm 0.09$ and a DICE score of $95.7\% \pm 0.09$. On the clinical testing set, it maintained strong performance with an mIoU score of $90.7\% \pm 0.01$ and a DICE score of $94.9\% \pm 0.01$, demonstrating its ability to generalize to real-world intraoral scans.

Conclusions: This study demonstrates the clinical potential of our AI-based modified multiview segmentation model for intraoral scans. The approach provides accurate results across varying scan qualities, supporting efficient digital model analysis in orthodontics and promising routine clinical use.

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1. Introduction

Accurate analysis of dental models is a critical component of modern orthodontic diagnosis and treatment planning. Increasingly,

digital dental models have become essential in clinical workflows, allowing orthodontists to evaluate occlusion, assess tooth positions, simulate treatment mechanics and lessen patient discomfort [1]. As the digital models continuously replace traditional

Abbreviations and Acronyms: 2D, Two-dimensional; 3D, Three-dimensional; AI, Artificial intelligence; CNN, Convolutional neural network; DGCNN, Dynamic graph convolutional neural network; DICE, Dice similarity coefficient; FDI, Fédération Dentaire Internationale (World Dental Federation); IOS, Intraoral scans; mIoU, Mean intersection over union; SDF, Shape diameter function; SGD, Stochastic gradient descent; TGNet, Tooth group network; TSegNet, Tooth segmentation network.

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plaster models, their clinical utility depends substantially on precision, efficiency, and reproducibility [2]. To perform model analysis with consistency and reliability, the digital workflow must begin with the precise identification and separation of individual teeth [3]. This foundational step ensures that relevant parameters such as tooth width, spacing or crowding can be accurately interpreted. As a result, the process of tooth segmentation becomes critical, serving as the gateway to a wide range of diagnostic and planning procedures in digital orthodontics [4].

Tooth segmentation is the process of isolating each individual tooth from a three-dimensional (3D) digital model [4]. It enables procedures such as space analysis, anchorage planning, tooth movement simulation, appliances fabrication including clear aligners, indirect bonding trays, and bracket customization [5]. Traditionally, the segmentation was performed manually and considered to be the gold standard. However, it requires significant time, effort, and operator expertise [6]. This labor-intensive process limits its scalability and practicality in routine practice. Barreto et al. reported that virtual setups are as reliable as manual setups while significantly reducing the time and effort required [7]. To fully leverage digital workflows, there is a pressing need for robust, fully automated segmentation methods that are accurate and generalizable across diverse clinical settings.

Artificial intelligence (AI) has introduced new opportunities for automating various diagnostic processes in orthodontics including cephalometric analysis [8], skeletal pattern classification [9], predicting the need for orthodontic extractions [10], cervical vertebra maturation stages (CVMS) classification [11], treatment outcome prediction [12] and soft tissue prediction [13]. Specifically for dental models, AI-driven segmentation of intraoral scans (IOS) has the potential to streamline the clinical workflow by reducing the time and effort required for diagnosis and treatment planning [14].

Recent approaches to tooth segmentation, particularly deep learning, fall into two main categories: geometric deep learning and multiview segmentation. Geometric approaches including PointNet [15], PointNet++ [16], DGCNN [17], TSegNet [18], PointTransformer [19] and ToothGroupNetwork (TGNet) [20], directly analyze 3D mesh data but face challenges related to irregular surface structures, varying resolution, and high computational demand [21]. These limitations affect not only performance but also feasibility in time-constrained clinical settings. Multiview segmentation, on the other hand, transforms 3D surfaces into multiple 2D projections from various angles. This approach allows the use of well-established 2D image segmentation models, including transformer-based models [22]. For instance, Mask2Former [23] excelled at this task because it analyzed the entire image context and understood how different parts are related to each other.

While the multiview approach had shown promise in tooth segmentation such as FiboSeg [20], some limitations still exist. Poor multiview image capture from occluded or undersurface angles can introduce errors in AI model training and predictions [24]. In addition, clinical 3D models are often acquired in inconsistent orientations. Maxillary and mandibular scans may be oriented in opposite directions depending on scanner settings, which complicates downstream processing tasks such as segmentation and landmark identification [25].

Therefore, clinicians require automated tooth segmentation solutions that address three critical unmet needs in orthodontic digital workflows: accuracy and robustness across varying scan qualities and resolutions, computational efficiency that enables rapid processing compatible with clinical timeframes, and minimal manual intervention. Existing approaches had not adequately balanced these requirements, creating a barrier to widespread adoption of automated segmentation in orthodontic practice.

Hence, a novel pipeline was developed to improve three key areas: (1) eliminating manual preprocessing and reducing operator dependency; (2) minimizing computational cost while maintaining comprehensive coverage of clinically relevant dental surfaces, and (3) ensuring compatibility across different intraoral scanner types. These contributions collectively advanced the feasibility of integrating automated tooth segmentation into routine orthodontic digital workflows, including treatment planning, virtual setup, and appliance design.

The objective of this study is to develop a multiview segmentation pipeline for IOS that is optimized for orthodontic applications. In addition, we evaluated our modified multiview segmentation approach by comparing it against geometric deep learning techniques, focusing on segmentation accuracy, computational efficiency, and generalizability across real-world clinical datasets.

2. Material and methods

This study protocol was approved by the Mahidol University Institutional Review Board, Faculty of Dentistry/Pharmacy, under the approval number COA.No.MU-DT/PY-IRB 2023/040.2107. Two datasets were used in this study:

2.1. Data source

Dataset 1—Training and initial validation

A total of 1200 IOS from the “3D Teeth Scan Segmentation and Labeling Challenge” were gathered (hereafter referred to as the TeethChallenge dataset) [20]. This publicly accessible dataset provided IOS with per-tooth segmentation labels. The study allocated 840 scans for training, 180 for validation, and 180 for testing. The *validation set* was used for monitoring the model's performance during training. The *testing set* was used to evaluate the model's segmentation performance after training.

Dataset 2—Clinical validation

A total of 29 anonymized IOS (14 maxillary and 15 mandibular) were randomly selected from the Department of Orthodontics, Faculty of Dentistry, Mahidol University (hereafter referred to as the MUOrtho dataset). These scans were obtained using the iTero Element 2 intraoral scanner (Align Technologies, San Jose, CA, USA). All clinical scans were deidentified prior to analysis to ensure patient privacy and compliance with ethical standards. No identifiable patient information was accessible to researchers during the study. The MUOrtho dataset was used to assess the generalization ability of the AI models and evaluate their segmentation performance after training.

The inclusion criteria were as follows: (1) IOS were anonymously collected from individuals over 12 years of age with complete permanent dentition defined as the presence of teeth from the first right molar to the first left molar. (2) IOS were free from scanning defects. (3) They exhibited mild to moderate crowding. The exclusion criteria included IOS with congenital dental anomalies, severe dental caries or tooth wear, congenital or acquired missing teeth, the presence of supernumerary teeth, and severe crowding.

2.2. Segmentation pipeline development

The overall developed tooth segmentation pipeline consisted of four main steps (Fig. 1): Orientation normalization (Fig. 1a), View-point generation (Fig. 1b), 2D segmentation (Fig. 1c), and the multiview fusion of 2D labels (Fig. 1d). The following subsections detailed the core components of the pipeline.

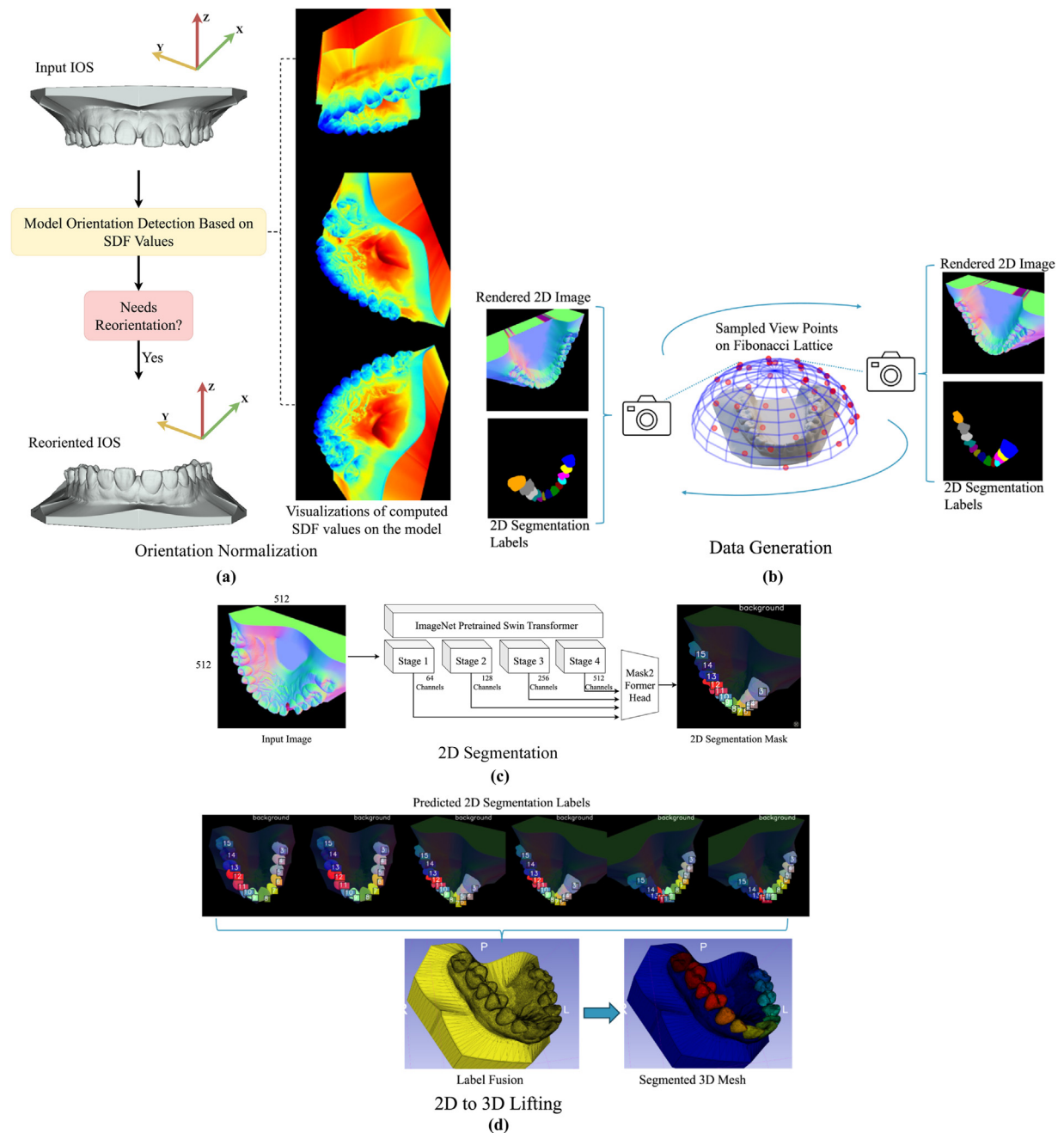


Fig. 1. The overall developed tooth segmentation pipeline consisted of four main steps: (a) Orientation Normalization Pipeline: Left picture show a scanned model of the upper arch, center picture show a visualization of computed shape diameter function (SDF) values (smaller values with cooler tones and the larger ones with warmer tone colors), right picture show the result of upper arch after automatic orientation detection and standardization. (b) Viewpoint generation was limited to the upper half of the sphere around the 3D model to capture clinically relevant regions and avoid unhelpful angles such as undersides of teeth with little clinical information. (c) Multiview 2D images were used to train a Mask2Former model within the MMSegmentation framework, utilizing a Swin-B Transformer backbone to extract features and generate per-tooth segmentation masks. (d) Predicted 2D segmentation labels from multiple views were fused to reconstruct a 3D surface, producing the final segmented 3D dental model.

2.2.1. IOS orientation normalization

To address the inconsistent orientation, we incorporated an orientation normalization step using the Shape Diameter Function (SDF) [26] to analyze the geometric properties of each scan and determine its proper orientation (Appendix A). The SDF assigns a thickness value to each vertex on the surfaces: lower values are found in slender regions such as cusp tips, while higher values are associated with thicker areas around the cervical area (Fig. 1a). After computing the SDF, two distinct sets of points were identified,

one representing thin regions (cusp tips) and the other representing thick regions (cervical tooth area).

By comparing the average vertical (z-axis) positions of these two groups, we could reliably distinguish upper from lower arches and detect whether the scan was inverted. For instance, in a raw maxillary scan with cusp tips facing downward, the thin regions (cusp tips) have lower vertical coordinates than the thick regions. Once the arch type and orientation were determined, each IOS was reoriented into a standardized pose: cusp tips aligned upward

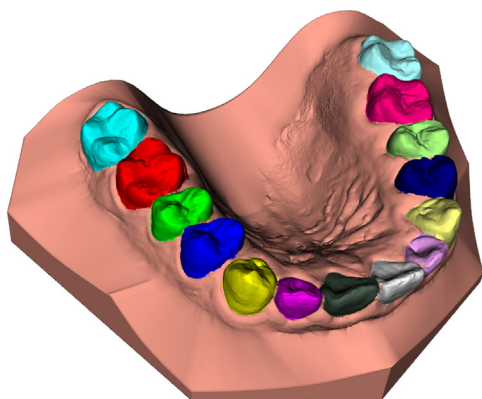


Fig. 2. Manual segmentation by brush mode in MeshLabler.

along the positive z-axis, and anterior teeth directed consistently forward. The full orientation normalization process is detailed in Algorithm 1 (Appendix B). This alignment step ensured consistent model positioning across the dataset and facilitated a more stable and accurate segmentation process.

2.2.2. Viewpoint generation

To ensure clinically relevant regions were well captured, our approach limited the generation of viewpoints to the upper half of the space surrounding the 3D model (Fig. 1b). This helped avoid unhelpful angles, such as those showing only the undersides of teeth, which often contain little or limited clinical information. Instead of selecting camera angles randomly, we used a Fibonacci lattice to evenly distribute viewpoints across the upper hemisphere of the 3D model (Fig. 1b). This ensured that the model was visualized from multiple well-balanced directions, capturing all important surfaces with consistent quality. We then used PyTorch3D [27], a rendering tool for 3D data, to generate 2D images from these viewpoints. Each 3D model was rendered into multiple 2D images, representing clinically meaningful perspectives with clear visualization of tooth anatomy. These 2D images were later used as input for the segmentation model in the next stage. By focusing on informative regions and avoiding poor-quality views, this strategy helped the model learn more effectively. It improved the consistency of segmentation outcomes and made the entire process faster and more reliable. Further technical details are provided in Appendix C.

2.2.3. 2D segmentation model

The 2D images generated from the previous step were used to train our model to automatically recognize and separate teeth in each image. To build and train this model, we used MMSegmentation, an open-source software toolkit designed for image segmentation tasks [28]. Within this framework, we selected Mask2Former, a segmentation model based on transformer architecture that has shown strong performance on complex images like tooth surfaces [23]. As illustrated in Fig. 1c, the Mask2Former model included three main parts. First, it extracted important visual features from the image. Then, it processed these features in stages to understand the spatial structure. Finally, it produced an output that marked the areas corresponding to teeth.

To help the model better understand patterns across each image, we used a transformer-based backbone network called Swin-B Transformer [29] to capture both fine details and broader spatial relationships. The 2D IOS images used in this stage were already generated in the previous step using PyTorch3D [27], which

converted each 3D model into a set of 2D images from various clinically relevant angles. These images served as direct inputs for model training. The model was trained to classify each pixel as either tooth or non-tooth using a combination of two loss functions: cross-entropy and the DICE scores, to improve training stability and segmentation accuracy.

2.2.4. Multiview fusion of 2D segmentations

To create a complete 3D segmentation from the 2D results, we combined predictions from all 2D views using the steps shown in Fig. 1d (see Appendix D for details). The details were as follows

- Back projection: Each 2D prediction was mapped back onto the surface of the original 3D model. This allowed us to transfer information from flat images to the full 3D shape.
- Fusion: All predictions from different angles were combined to form a single complete 3D result.

This process produced a fully segmented IOS in which each component of the model, including individual teeth and gingiva, was clearly identified. This completed the segmentation pipeline.

2.3. Data annotation for evaluation

All 29 scans in the MUOrtho dataset were manually segmented into individual teeth and gingiva to establish ground truth labels for evaluation. Segmentation was performed using MeshLabeler (Tai-Hsien Wu, open-source annotation tool) [30], where each tooth was delineated as a distinct region with a specific label (Fig. 2). Two experienced clinicians (SN, NS) independently carried out the labeling process for all scans. To assess consistency between annotators, the segmentation of each scan produced by both clinicians was compared for agreement. Inter-annotator reliability was quantified using Cohen's kappa statistic for each scan and Intersection over Union (IoU) scores were subsequently computed between each pair of annotated scans.

2.4. Training and inference setup

For training the 2D segmentation model, 40 2D views of each IOS model with a resolution of 512×512 were generated using our view generation strategy. The generated 2D segmentation dataset for training contained 33,600 (840 scans × 40 views) images and segmentation masks. We trained the Mask2Former model using Stochastic Gradient Descent (SGD) with momentum 0.9 and weight decay 0.0005, an initial learning rate of 0.01, 20,000 training iterations, and a batch size of 4. The loss function combined cross-entropy and DICE loss:

$$L = \omega_1 L_{CE} + \omega_2 L_{Dice}$$

where ω_1 and ω_2 are weighting factors, L_{CE} is the cross-entropy loss, and L_{Dice} is the DICE loss. The training was done on a single RTX 3090 GPU with 24GB of memory.

Inference time was evaluated on 5 randomly selected clinical scans to assess the computational feasibility of the proposed method for clinical deployment. Using hardware consisting of an NVIDIA RTX 2080ti GPU (11GB VRAM), AMD Ryzen 9 3900x CPU, and 64GB RAM, the end-to-end processing time for each stage of the segmentation pipeline was measured. The evaluation included all processing stages: data loading, multiview rendering (40 views), model inference, aggregation, and postprocessing. Each scan was processed three times, and the average timing was recorded.

2.5. Vertex distribution of datasets

To characterize the complexity and resolution of the datasets, we analyzed the vertex distribution across all IOS scans. Higher vertex counts indicate higher-resolution scans that capture finer anatomical details but require greater computational resources. The number of vertices in each scan was computed by a Python program (Python Software Foundation, Wilmington, DE, USA) and distributions were presented using histograms. This analysis provided insights into scan resolution variability between the TeethChallenge and MUOrtho datasets which directly impacted computational requirements and segmentation precision.

2.6. Quantitative evaluation of the segmentation performance

The segmentation performance of the algorithms on the IOS datasets was quantitatively evaluated using two standard metrics: IoU and DICE scores. These metrics measure the spatial overlap between the predicted segmentation (P) and the ground truth (G) by comparing the sets of mesh faces assigned to each tooth label.

For each tooth class c (where c corresponds to individual teeth classes identified using the Fédération Dentaire Internationale (FDI) numbering system: 11–18, 21–28, 31–38, 41–48), we define P_c as the set of predicted mesh faces belonging to tooth class c , and G_c as the corresponding set of ground truth faces for that tooth.

IoU, also known as the Jaccard index, measures the ratio of the intersection to the union of the predicted and ground-truth sets for a given class c . It is calculated as:

$$IoU_c = \frac{|P_c \cap G_c|}{|P_c \cup G_c|} \quad (1)$$

where P_c represents the set of faces predicted as class c , G_c is the set of ground truth faces for class c in the set. A higher IoU value indicates better overlap, with 1 representing perfect agreement.

In contrast, the DICE scores also measure the overlap but normalizes it by averaging the size of the prediction and ground-truth sets rather than taking their union. This difference makes DICE more sensitive to boundaries of the structures. For this reason, DICE is often reported alongside IoU and is defined for a tooth class c as:

$$DSC_c = \frac{2|P_c \cap G_c|}{|P_c| + |G_c|} \quad (2)$$

DSC ranges from 0 (no overlap) to 1 (perfect overlap).

These metrics were calculated on a per-tooth basis for each individual tooth identified in the scan. To obtain an overall performance measure for tooth segmentation on each scan, the mIoU and mean DICE scores were computed by averaging the respective per-tooth scores across all valid tooth labels present in that specific scan:

$$mIoU = \frac{1}{|C_{valid_teeth}|} \sum_{c \in C_{valid_teeth}} IoU_c \quad (3)$$

$$Mean\ DICE = \frac{1}{|C_{valid_teeth}|} \sum_{c \in C_{valid_teeth}} DICE_c \quad (4)$$

where C_{valid_teeth} represents the set of tooth classes present in the specific ground truth or prediction for that sample.

Furthermore, to assess our model's performance on MUOrtho dataset, individual tooth labels were categorized into the following classes: Central Incisor, Lateral Incisor, Canine, Premolar, First Molar, Second Molar and Third Molar. Aggregated scores for each tooth class were computed by averaging the per-tooth DICE and IoU scores for all individual teeth belonging to the respective tooth

class in the evaluated scan. For example, the average DICE for the Third Molar category was calculated as the mean DICE value across all third molar teeth (FDI labels 18, 28, 38, 48) present in the evaluation results for that scan. Overall performance across the entire dataset was determined by averaging these calculated metrics (mIoU, Mean DICE, and per-tooth aggregated scores) over all IOS.

In this study, we compared the performance of our modified multiview Mask2Former approach with six existing 3D segmentation geometric deep learning-based models, including PointNet [15], PointNet++ [16], Dynamic Graph Convolution CNN (DGCNN) [17], TSegNet [18], PointTransformer [19], and ToothGroupNetwork (TGNet) [20]. Both the geometric models and our multiview approach underwent evaluation on the TeethChallenge testing set and the MUOrtho dataset to ensure a comprehensive assessment. We performed paired Wilcoxon signed-rank tests to compare per-scan mIoU and DICE scores between the multiview Mask2Former and each baseline method. The nonparametric test was chosen because the Shapiro-Wilk tests demonstrated non-normal data distributions. To visualize the segmentation results, we captured 2D renderings of a randomly sampled MUOrtho datasets from occlusal and lateral views after segmentation, allowing for qualitative assessment of the segmentation quality and boundary delineation accuracy across different approaches.

3. Results

3.1. Segmentation performance on the TeethChallenge testing set

The training and testing sets of the TeethChallenge dataset were used to train and test the segmentation models. Tables 1 and 2 present the DICE and mIoU metrics for the tooth classes on this testing set. Specifically, our multiview Mask2Former model achieved the following DICE scores on the TeethChallenge testing set: gingiva (0.971 ± 0.02) central incisors (0.966 ± 0.02), lateral incisors (0.964 ± 0.03), canines (0.964 ± 0.04), premolars (0.958 ± 0.05), first molars (0.959 ± 0.11), second molars (0.895 ± 0.44), and third molars (0.829 ± 0.00). The corresponding mIoU scores were: 0.943 ± 0.03 , 0.935 ± 0.03 , 0.933 ± 0.04 , 0.934 ± 0.07 , 0.934 ± 0.08 , 0.940 ± 0.13 , 0.867 ± 0.43 and 0.741 ± 0.00 respectively. Overall performance metrics showed mIoU score of 0.931 ± 0.09 and DICE score of 0.957 ± 0.09 (Table 3). Compared to baseline methods, our approach outperformed PointNet (DICE: (0.843 ± 0.11)), PointNet++ (0.923 ± 0.09), DGCNN (0.878 ± 0.09), TSegNet (0.905 ± 0.10) and PointTransformer (0.934 ± 0.09). Although TGNet achieved the highest overall DICE (0.963 ± 0.08) and mIoU (0.944 ± 0.09) scores, our approach was comparable in accuracy (DICE: 0.957 ± 0.09 and mIoU: 0.931 ± 0.09 scores), and notably excelled at third molar segmentation, a particularly challenging tooth class. Wilcoxon signed-rank tests on per-scan mIoU and DICE scores data confirmed statistically significant differences between the multiview Mask2Former and all baseline methods ($P < 0.01$ for all comparisons).

3.2. Inter-annotator agreement for MUOrtho dataset

The manual annotations showed excellent inter-rater reliability. Cohen's Kappa values ranged from 0.9901 to 0.9999 across the 29 scans, with a mean of 0.9968 (SD = 0.01). The mean IoU between annotators was 0.9953 (SD = 0.01). These near-perfect agreement scores validate the quality of our ground truth annotations.

Table 1

Per class averaged DICE scores of intraoral scan segmentation models on TeethChallenge testing set.

Model	Gingiva	Central incisor	Lateral incisor	Canine	Premolar	First molar	Second molar	Third molar
Multiview Mask2Former ($\pm$ SD)	0.971 ($\pm$ 0.02)	0.966 ($\pm$ 0.02)	0.964 ($\pm$ 0.03)	0.964 ($\pm$ 0.04)	0.958 ($\pm$ 0.05)	0.959 ($\pm$ 0.11)	0.895 ($\pm$ 0.44)	0.829 ($\pm$ 0.00)
TGNet ($\pm$ SD)	0.978 ($\pm$ 0.01)	0.970 ($\pm$ 0.03)	0.967 ($\pm$ 0.06)	0.969 ($\pm$ 0.05)	0.966 ($\pm$ 0.04)	0.961 ($\pm$ 0.09)	0.905 ($\pm$ 0.44)	0.489 ($\pm$ 0.07)
Point transformer ($\pm$ SD)	0.965 ($\pm$ 0.01)	0.947 ($\pm$ 0.04)	0.950 ($\pm$ 0.05)	0.943 ($\pm$ 0.05)	0.934 ($\pm$ 0.06)	0.944 ($\pm$ 0.12)	0.826 ($\pm$ 0.44)	0.109 ($\pm$ 0.02)
TSegNet ($\pm$ SD)	0.947 ($\pm$ 0.03)	0.856 ($\pm$ 0.13)	0.860 ($\pm$ 0.18)	0.911 ($\pm$ 0.11)	0.930 ($\pm$ 0.07)	0.938 ($\pm$ 0.13)	0.870 ($\pm$ 0.44)	0.000 ($\pm$ 0.00)
DGCNN ($\pm$ SD)	0.952 ($\pm$ 0.02)	0.919 ($\pm$ 0.04)	0.911 ($\pm$ 0.08)	0.895 ($\pm$ 0.09)	0.895 ($\pm$ 0.06)	0.908 ($\pm$ 0.13)	0.626 ($\pm$ 0.41)	0.005 ($\pm$ 0.00)
PointNet++ ($\pm$ SD)	0.961 ($\pm$ 0.02)	0.942 ($\pm$ 0.04)	0.940 ($\pm$ 0.06)	0.930 ($\pm$ 0.07)	0.932 ($\pm$ 0.05)	0.934 ($\pm$ 0.14)	0.779 ($\pm$ 0.43)	0.000 ($\pm$ 0.00)
PointNet ($\pm$ SD)	0.941 ($\pm$ 0.02)	0.901 ($\pm$ 0.06)	0.878 ($\pm$ 0.11)	0.857 ($\pm$ 0.13)	0.852 ($\pm$ 0.07)	0.870 ($\pm$ 0.14)	0.596 ($\pm$ 0.40)	0.000 ($\pm$ 0.00)

Table 2

Per class averaged mIoU scores of intraoral scan segmentation models on TeethChallenge testing set.

Model	Gingiva	Central incisor	Lateral incisor	Canine	Premolar	First molar	Second molar	Third molar
Multiview Mask2Former ($\pm$ SD)	0.943 ($\pm$ 0.03)	0.935 ($\pm$ 0.03)	0.933 ($\pm$ 0.04)	0.934 ($\pm$ 0.07)	0.934 ($\pm$ 0.08)	0.940 ($\pm$ 0.13)	0.867 ($\pm$ 0.43)	0.741 ($\pm$ 0.00)
TGNet ($\pm$ SD)	0.958 ($\pm$ 0.03)	0.948 ($\pm$ 0.05)	0.946 ($\pm$ 0.08)	0.946 ($\pm$ 0.08)	0.950 ($\pm$ 0.07)	0.948 ($\pm$ 0.12)	0.879 ($\pm$ 0.43)	0.478 ($\pm$ 0.07)
Point transformer ($\pm$ SD)	0.933 ($\pm$ 0.03)	0.907 ($\pm$ 0.05)	0.910 ($\pm$ 0.07)	0.907 ($\pm$ 0.07)	0.902 ($\pm$ 0.08)	0.918 ($\pm$ 0.15)	0.792 ($\pm$ 0.42)	0.070 ($\pm$ 0.01)
TSegNet ($\pm$ SD)	0.901 ($\pm$ 0.05)	0.807 ($\pm$ 0.17)	0.812 ($\pm$ 0.20)	0.869 ($\pm$ 0.14)	0.896 ($\pm$ 0.10)	0.913 ($\pm$ 0.15)	0.828 ($\pm$ 0.42)	0.000 ($\pm$ 0.00)
DGCNN ($\pm$ SD)	0.909 ($\pm$ 0.02)	0.857 ($\pm$ 0.04)	0.847 ($\pm$ 0.08)	0.841 ($\pm$ 0.09)	0.840 ($\pm$ 0.06)	0.859 ($\pm$ 0.13)	0.576 ($\pm$ 0.41)	0.002 ($\pm$ 0.00)
PointNet++ ($\pm$ SD)	0.924 ($\pm$ 0.03)	0.897 ($\pm$ 0.05)	0.895 ($\pm$ 0.08)	0.890 ($\pm$ 0.09)	0.895 ($\pm$ 0.08)	0.905 ($\pm$ 0.16)	0.740 ($\pm$ 0.41)	0.000 ($\pm$ 0.00)
PointNet ($\pm$ SD)	0.889 ($\pm$ 0.04)	0.831 ($\pm$ 0.09)	0.801 ($\pm$ 0.14)	0.786 ($\pm$ 0.16)	0.780 ($\pm$ 0.10)	0.799 ($\pm$ 0.18)	0.531 ($\pm$ 0.37)	0.000 ($\pm$ 0.00)

Table 3

Per dataset averaged overall performance metrics (mIoU, DICE) scores of intraoral scan segmentation models on TeethChallenge testing set.

Model	mIoU	P-value	DICE	P-value
Multiview Mask2Former ($\pm$ SD)	0.931 ($\pm$ 0.09)	-	0.957 ($\pm$ 0.09)	-
TGNet ($\pm$ SD)	0.944 ($\pm$ 0.09)	<0.01*	0.963 ($\pm$ 0.08)	<0.01*
Point transformer ($\pm$ SD)	0.900 ($\pm$ 0.10)	<0.01*	0.934 ($\pm$ 0.09)	<0.01*
TSegNet ($\pm$ SD)	0.866 ($\pm$ 0.11)	<0.01*	0.905 ($\pm$ 0.10)	<0.01*
DGCNN ($\pm$ SD)	0.823 ($\pm$ 0.11)	<0.01*	0.878 ($\pm$ 0.09)	<0.01*
PointNet++ ($\pm$ SD)	0.884 ($\pm$ 0.10)	<0.01*	0.923 ($\pm$ 0.09)	<0.01*
PointNet ($\pm$ SD)	0.774 ($\pm$ 0.13)	<0.01*	0.843 ($\pm$ 0.11)	<0.01*

*P-value <0.01 from Wilcoxon signed-rank test on per-scan paired data.

Performance comparison of intraoral scan segmentation models on the TeethChallenge testing set.

3.3. Distribution of the number of vertices in the datasets

The TeethChallenge dataset showed mean vertex distributions around 110,000 vertices per scan (Fig. 3a and b). In contrast, the MUOrtho dataset exhibited significantly higher resolution, averaging approximately 200,000 vertices per scan (Fig. 3c). This two-fold difference in vertex density demonstrates the resolution gap between TeethChallenge datasets and real-world clinical scans, presenting a significant challenge for model generalization.

3.4. Segmentation performance on the MUOrtho datasets

On the MUOrtho dataset, our approach achieved DICE scores of: gingiva (0.967 ± 0.02), central incisors (0.940 ± 0.02), lateral incisors (0.940 ± 0.01), canines (0.946 ± 0.01), premolars (0.948 ± 0.01), first molars (0.963 ± 0.01), second molars (0.964 ± 0.01), and third molars (0.757 ± 0.33) (Table 4). The mIoU metrics were: 0.937 ± 0.03 , 0.889 ± 0.03 , 0.890 ± 0.02 , 0.899 ± 0.02 , 0.905 ± 0.01 , 0.929 ± 0.01 , 0.931 ± 0.02 , and 0.719 ± 0.31 respectively (Table 5). Overall clinical performance showed mIoU score of 0.907 ± 0.01 and DICE score of 0.949 ± 0.01 (Table 6), surpassing all baseline methods including TGNet (DICE: 0.819 ± 0.07 , mIoU: 0.733 ± 0.11). Wilcoxon signed-rank tests on per-scan mIoU and DICE scores data confirmed statistically significant differences between the multiview Mask2Former and all baseline methods ($P < 0.01$ for all comparisons). This demonstrates superior generalization to real-world clinical data.

3.5. Inference time for segmentation

The average end-to-end inference time was 23.71 ± 2.23 seconds per scan ($n = 5$) on the specified hardware configuration (RTX 2080ti GPU, Ryzen 9 3900x CPU, 64GB RAM). The processing pipeline time was distributed as follows: data loading (0.64 seconds, 2.7%), rendering 40 views (6.70 seconds, 28.3%), model inference (1.90 seconds, 8.0%), multiview aggregation (0.02 seconds, 0.1%), and postprocessing (14.44 seconds, 60.9%).

3.6. Qualitative comparison

Visual analysis revealed distinct advantages of our multi-view approach over geometric approaches (Fig. 4). Our segmentations produced smoother and more anatomically accurate boundaries, especially in challenging regions such as gingival margins and posterior teeth. In these areas, our approach correctly delineated cervical boundaries and interproximal contacts while preserving clear separation between adjacent teeth whereas geometric approaches often yielded fragmented or merged segments.

The other geometric approaches frequently generated irregular contours with visible artifacts, including jagged edges along tooth boundaries, incorrect merging of adjacent teeth in crowded areas and inconsistent gingival segmentation.

4. Discussion

Precise identification and separation of individual teeth are essential for orthodontic diagnosis and treatment planning, outcome

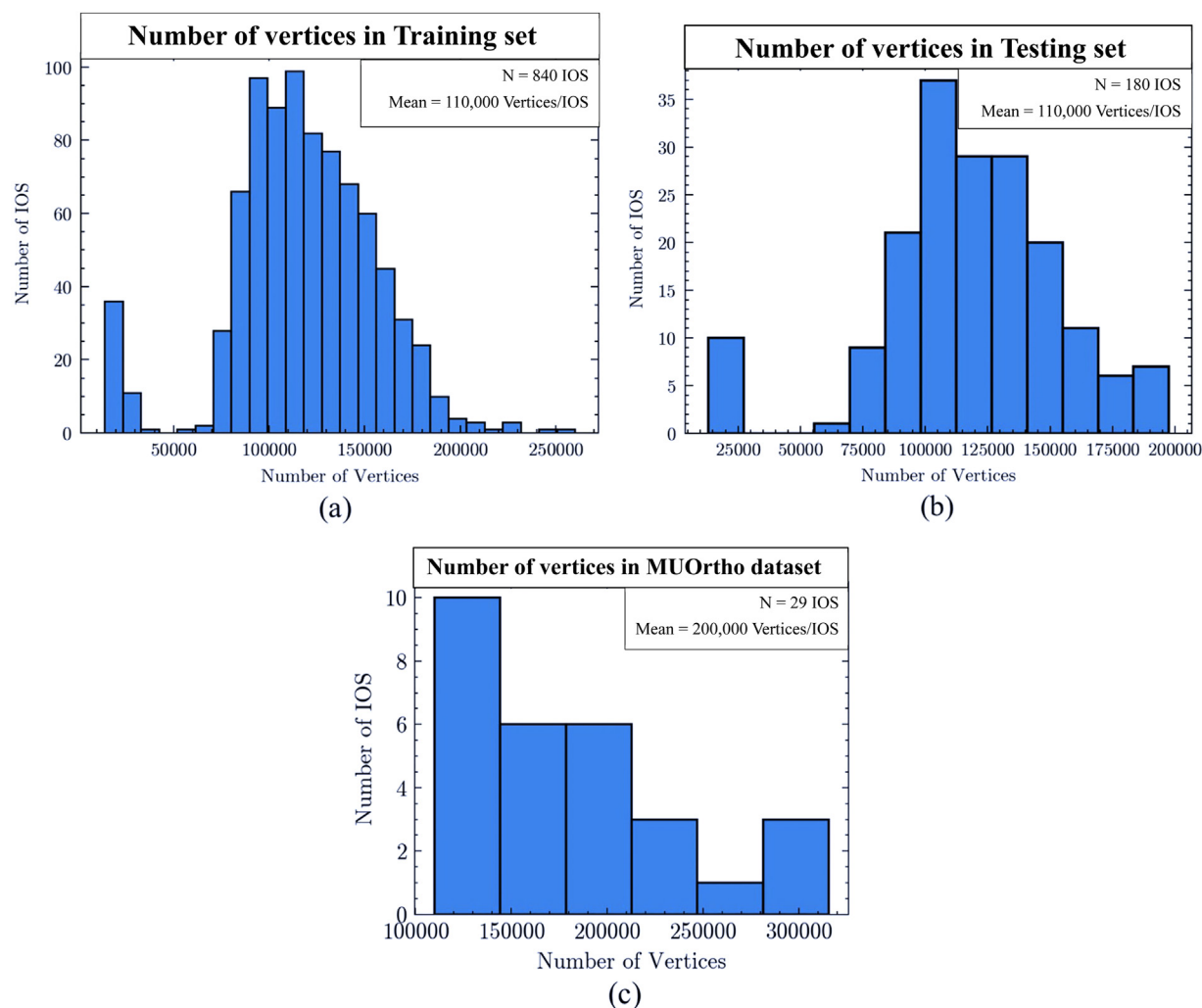


Fig. 3. Distribution of the number of vertices contained in different IOS datasets: (a) TeethChallenge training set (b) TeethChallenge testing set (c) MUOrtho dataset.

Table 4
Per class averaged DICE scores of intraoral scan segmentation models on MUOrtho dataset.

Model	Gingiva	Central incisor	Lateral incisor	Canine	Premolar	First molar	Second molar	Third molar
Multiview Mask2Former ($\pm$ SD)	0.967 ($\pm$ 0.02)	0.940 ($\pm$ 0.02)	0.940 ($\pm$ 0.01)	0.946 ($\pm$ 0.01)	0.948 ($\pm$ 0.01)	0.963 ($\pm$ 0.01)	0.964 ($\pm$ 0.01)	0.757 ($\pm$ 0.33)
TGNet ($\pm$ SD)	0.953 ($\pm$ 0.02)	0.848 ($\pm$ 0.09)	0.838 ($\pm$ 0.11)	0.844 ($\pm$ 0.09)	0.785 ($\pm$ 0.07)	0.827 ($\pm$ 0.14)	0.786 ($\pm$ 0.22)	0.372 ($\pm$ 0.23)
Point transformer ($\pm$ SD)	0.956 ($\pm$ 0.02)	0.926 ($\pm$ 0.03)	0.920 ($\pm$ 0.03)	0.931 ($\pm$ 0.03)	0.907 ($\pm$ 0.04)	0.886 ($\pm$ 0.11)	0.835 ($\pm$ 0.15)	0.000 ($\pm$ 0.00)
TSegNet ($\pm$ SD)	0.911 ($\pm$ 0.04)	0.535 ($\pm$ 0.32)	0.481 ($\pm$ 0.36)	0.673 ($\pm$ 0.33)	0.821 ($\pm$ 0.10)	0.805 ($\pm$ 0.20)	0.646 ($\pm$ 0.33)	0.000 ($\pm$ 0.00)
DGCNN ($\pm$ SD)	0.942 ($\pm$ 0.02)	0.764 ($\pm$ 0.11)	0.716 ($\pm$ 0.19)	0.742 ($\pm$ 0.18)	0.729 ($\pm$ 0.11)	0.665 ($\pm$ 0.23)	0.572 ($\pm$ 0.33)	0.001 ($\pm$ 0.00)
PointNet++ ($\pm$ SD)	0.947 ($\pm$ 0.03)	0.897 ($\pm$ 0.05)	0.889 ($\pm$ 0.06)	0.901 ($\pm$ 0.05)	0.886 ($\pm$ 0.04)	0.887 ($\pm$ 0.10)	0.836 ($\pm$ 0.18)	0.000 ($\pm$ 0.00)
PointNet ($\pm$ SD)	0.929 ($\pm$ 0.03)	0.796 ($\pm$ 0.10)	0.599 ($\pm$ 0.28)	0.573 ($\pm$ 0.34)	0.548 ($\pm$ 0.16)	0.610 ($\pm$ 0.26)	0.567 ($\pm$ 0.36)	0.000 ($\pm$ 0.00)

Table 5
Per class averaged mIoU scores of intraoral mesh segmentation models on MUOrtho dataset.

Model	Gingiva	Central incisor	Lateral incisor	Canine	Premolar	First molar	Second molar	Third molar
Multiview Mask2Former ($\pm$ SD)	0.937 ($\pm$ 0.03)	0.889 ($\pm$ 0.03)	0.890 ($\pm$ 0.02)	0.899 ($\pm$ 0.02)	0.905 ($\pm$ 0.01)	0.929 ($\pm$ 0.01)	0.931 ($\pm$ 0.02)	0.719 ($\pm$ 0.31)
TGNet ($\pm$ SD)	0.911 ($\pm$ 0.04)	0.751 ($\pm$ 0.13)	0.739 ($\pm$ 0.14)	0.751 ($\pm$ 0.13)	0.687 ($\pm$ 0.11)	0.741 ($\pm$ 0.18)	0.729 ($\pm$ 0.25)	0.348 ($\pm$ 0.19)
Point transformer ($\pm$ SD)	0.916 ($\pm$ 0.04)	0.864 ($\pm$ 0.06)	0.853 ($\pm$ 0.05)	0.873 ($\pm$ 0.05)	0.842 ($\pm$ 0.06)	0.819 ($\pm$ 0.15)	0.772 ($\pm$ 0.20)	0.000 ($\pm$ 0.00)
TSegNet ($\pm$ SD)	0.839 ($\pm$ 0.07)	0.495 ($\pm$ 0.31)	0.440 ($\pm$ 0.33)	0.630 ($\pm$ 0.33)	0.762 ($\pm$ 0.14)	0.742 ($\pm$ 0.23)	0.613 ($\pm$ 0.33)	0.000 ($\pm$ 0.00)
DGCNN ($\pm$ SD)	0.892 ($\pm$ 0.04)	0.656 ($\pm$ 0.14)	0.615 ($\pm$ 0.22)	0.657 ($\pm$ 0.22)	0.627 ($\pm$ 0.15)	0.565 ($\pm$ 0.26)	0.493 ($\pm$ 0.33)	0.001 ($\pm$ 0.00)
PointNet++ ($\pm$ SD)	0.900 ($\pm$ 0.05)	0.818 ($\pm$ 0.07)	0.805 ($\pm$ 0.08)	0.826 ($\pm$ 0.08)	0.811 ($\pm$ 0.07)	0.817 ($\pm$ 0.14)	0.759 ($\pm$ 0.21)	0.000 ($\pm$ 0.00)
PointNet ($\pm$ SD)	0.868 ($\pm$ 0.04)	0.675 ($\pm$ 0.13)	0.488 ($\pm$ 0.27)	0.482 ($\pm$ 0.32)	0.449 ($\pm$ 0.20)	0.496 ($\pm$ 0.27)	0.487 ($\pm$ 0.34)	0.000 ($\pm$ 0.00)

Table 6

Per dataset averaged overall performance metrics (mIoU, DICE) scores of intraoral scan segmentation models on MUOrtho dataset.

Model	mIoU	P-value	DICE	P-value
Multiview Mask2Former ($\pm$ SD)	0.907 ($\pm$ 0.01)	-	0.949 ($\pm$ 0.01)	-
TGNet ($\pm$ SD)	0.733 ($\pm$ 0.11)	<0.01*	0.819 ($\pm$ 0.07)	<0.01*
Point transformer ($\pm$ SD)	0.830 ($\pm$ 0.06)	<0.01*	0.892 ($\pm$ 0.04)	<0.01*
TSegNet ($\pm$ SD)	0.640 ($\pm$ 0.14)	<0.01*	0.689 ($\pm$ 0.10)	<0.01*
DGCNN ($\pm$ SD)	0.611 ($\pm$ 0.15)	<0.01*	0.702 ($\pm$ 0.11)	<0.01*
PointNet++ ($\pm$ SD)	0.801 ($\pm$ 0.07)	<0.01*	0.874 ($\pm$ 0.04)	<0.01*
PointNet ($\pm$ SD)	0.514 ($\pm$ 0.20)	<0.01*	0.611 ($\pm$ 0.16)	<0.01*

*P-value <0.01 from Wilcoxon signed-rank on per-scan paired data.

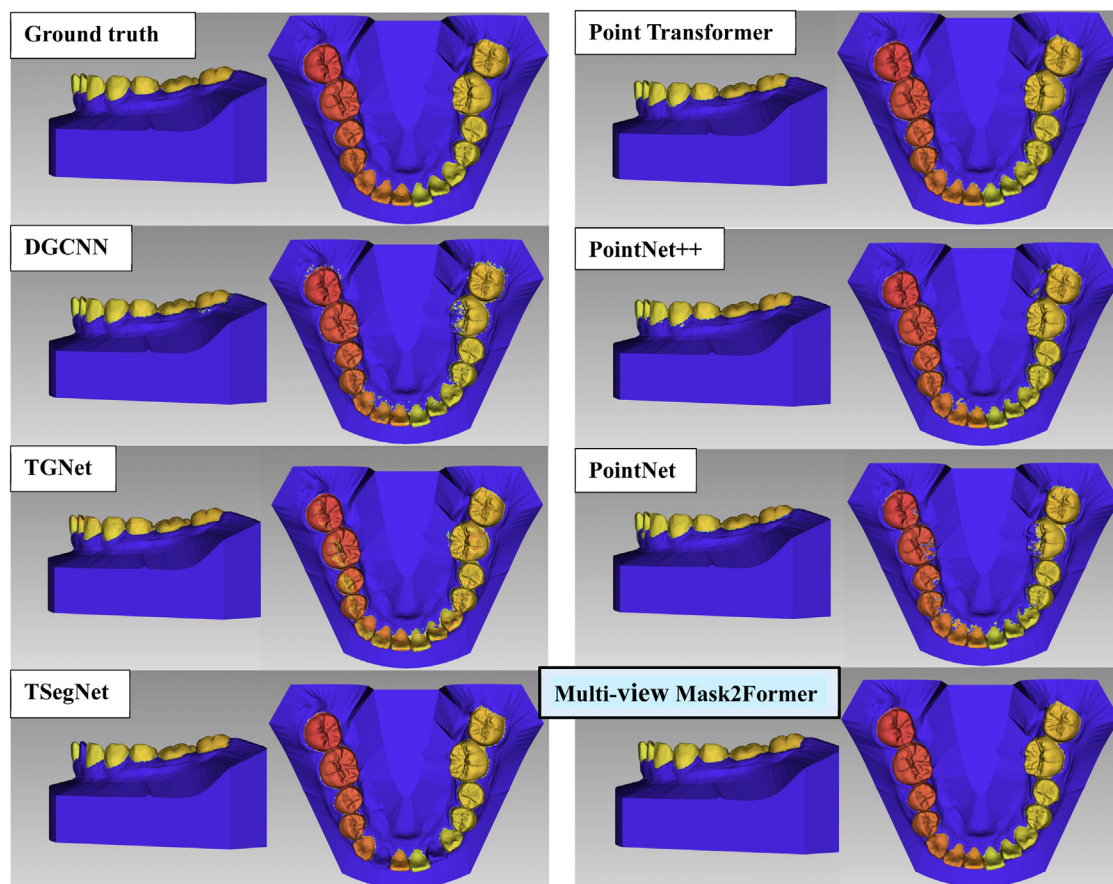


Fig. 4. Qualitative comparison of segmentation results on the test set. Each color gradient represents a specific tooth type, while the blue region indicates nondental structures. The first row on left column shows the ground truth reference. Subsequent rows present outputs from DGCNN, TGNet, TSegNet, Point Transformer, PointNet++, PointNet, and the proposed multiview Mask2Former. The multiview approach demonstrates fewer artifacts along tooth boundaries and provides more anatomically accurate segmentation of gingival and interproximal regions compared to geometric-based methods.

simulation, and appliance design. This study presented a novel automated segmentation approach that improved the identification of individual teeth and gingiva. Our modified multiview pipeline, utilizing the transformer-based Mask2Former model, demonstrated strong performance across both public benchmark data (TeethChallenge testing sets) and real-world clinical scans (MUOrtho dataset). On the TeethChallenge testing set, our approach achieved comparable or superior results to other well-established geometric segmentation models including DGCNN [17], TSegNet [18], Point Transformer [19], PointNet++ [16], and PointNet [15]. While TGNet [20] achieved the highest scores overall (Table 3), our approach demonstrated comparable accuracy, ranking as the second-best overall. It achieved the top performance for the third molar category, which is often challenging to segment due to its variable position and

anatomy. Notably, all approaches except TGNet failed to correctly identify this tooth group, highlighting the robustness of our approach.

The strength of the approach became more evident on the real-world clinical (MUOrtho) dataset, where scan conditions such as vertex density and anatomical complexity varied significantly from the training dataset. In this dataset, our approach demonstrated the highest segmentation accuracy, with DICE scores exceeding 0.94 for most tooth classes, including posterior teeth (Table 4). These variations in vertex density underscore the resolution gap between curated benchmark datasets and real-world clinical scans, a key challenge for model generalization. Nevertheless, our approach showed significantly higher mIoU and DICE scores and demonstrated robust performance across scans with varying resolutions and vertex

distributions. These results suggest that the approach is well-suited for clinical applications, where such variability is both common and inevitable.

Qualitative analysis (Fig. 4) showed that the segmentations produced by our approach were anatomically accurate and closely matched expert annotations, particularly in the gingival and interproximal regions. In contrast, geometric models often produced irregular contours or artifacts near tooth boundaries. This visual accuracy is clinically important, as accurate boundary definition is crucial for orthodontic treatment planning as well as appliance design. The errors in these areas could affect appliance fitting, appliance boundaries, and tooth movement prediction.

Three key technical innovations contributed to our approach. First, the automated orientation normalization using SDF [26] ensured consistent scan positioning regardless of initial orientation, addressing a common source of variability in clinical practice. Second, the Fibonacci lattice viewpoint sampling strategy provided comprehensive surface coverage while maintaining computational efficiency, capturing all clinically important surfaces, including challenging interproximal and occlusal areas. Third, reformulating 3D segmentation as multiple 2D segmentation tasks enabled us to leverage transformer-based models like Mask2Former [23], which excel at capturing both local details and global context in medical imaging. This strategy effectively handled the non-uniform vertex density that often challenges geometric models [21].

The computational performance analysis revealed that the proposed method achieved complete tooth segmentation in approximately 24 seconds per scan, representing a substantial time reduction compared to manual segmentation, which could take up to 2 hours per case [31]. This processing time was clinically feasible for integration into orthodontic workflows, particularly for treatment planning sessions where digital model analysis could be immediately performed after scan acquisition. The framework's ability to handle varying scan resolutions without down-sampling represents a significant clinical advantage, as it preserves anatomical detail critical for accurate treatment planning. However, the multiview approach does require additional computational steps including orientation correction, rendering, and fusion, which increases processing time compared to direct geometric approaches.

Additionally, while our viewpoint generation strategy proved effective across our testing set, optimal viewpoint selection may need adjustment for extreme clinical presentations such as severe crowding or extensive edentulous regions. These challenging clinical conditions represent a limitation of the current methodology. Another important limitation of this study was the lack of systematic tracking of manual correction requirements. While our quantitative metrics (mIoU, DICE) demonstrated high segmentation accuracy, the practical burden of postsegmentation editing, including correction time, frequency of required interventions, and types of errors requiring manual refinement, was not systematically tracked.

Therefore, future validation studies should incorporate formal tracking of correction time and error patterns to provide a complete picture of clinical applicability and to identify specific scenarios where the model required improvement. Future research should also explore several promising directions. Incorporating uncertainty estimation could guide clinicians to areas requiring manual review, particularly in complex cases. Validation on more challenging clinical scenarios, including mixed dentition, severe malocclusions, craniofacial anomalies such as cleft lip and palate, syndromes affecting dental development, mutilated dentitions, and extensive dental restorations would confirm the model's robustness and generalizability across a wide range of orthodontic and inter-

disciplinary applications and further expand its clinical utility. Additionally, developing adaptive viewpoint selection strategies based on initial scan assessment could optimize performance for specific clinical presentations while maintaining computational efficiency.

5. Conclusion

This multiview AI segmentation pipeline provided practical benefits in orthodontic practice. The approach achieved high accuracy and performance on both public and clinical datasets. It demonstrated robust generalization across varying resolutions and anatomical complexity. Additionally, the processing time was significantly reduced compared to manual segmentation, allowing for clinical adoption. These findings establish a practical foundation for further clinical development and use of segmentation tools in digital orthodontic workflows, including virtual setup, clear aligner design, and automated space analysis using deep learning techniques.

Availability of data and material

The datasets generated and/or analyzed during the current study are available at this url: https://drive.google.com/drive/u/0/folders/15oPOCZM_O_-Bir18VbSM8wRUEzoyLXby

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejwf.2025.12.002.

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